The bound $A(18,6,6) \le 183$

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Notation. $A(n,d,w)$ is the maximum size of a binary code of length n, constant weight w and minimum distance at least d. Throughout, $n = 18$, $d = 6$ and $w = 6$. A *word* is a 6-subset of $[18] = \{1,\dots,18\}$; there are $V = \binom{18}{6} = 18564$ words. Two words u, u' are at distance $2(6 - |u \cap u'|)$. So a *code* is a set of words any two of which meet in at most 3 points. A set of words is called *compatible* if it is a code.

Theorem. $A(18,6,6) \le 183$.

The bounds known before were $133 \le A(18,6,6) \le 186$, see [4]. The upper bound 186 follows from $A(17,6,6) \le 124$ (Östergård [7]) and the Johnson bound $A(18,6,6) \le \lfloor 18 \cdot 124/12 \rfloor$.

Outline. The proof is a semidefinite programming bound. Sections 1–6 give a list of linear and semidefinite constraints on certain numbers $z(S)$ attached to a code, and prove that every code satisfies them. Section 7 describes a certificate, in rational numbers, that these constraints imply $|C| < 184$; it is checked by computer in exact arithmetic. The constraints are those of the four-point bound of Polak [8] together with further constraints obtained by counting. Adding such constraints to a bound of this kind goes back to [1] and [2, Theorem 22]. Among them are 23 inequalities that come from the leaves of packings of 4-sets (Sections 3–5); such leaves are classical, see [3].

What is proved, and how it is checked. The Theorem rests on three things: (i) the constraints are valid for every code, which is proved in Sections 1–6; (ii) the certificate of Section 7, which is verified by the public program in exact arithmetic; (iii) the 16387 matrices in the public model file are correct instances of these constraints. For (iii), every matrix in the model carries a label naming its family and its parameters, and a written specification states how the coefficients follow from the label. All matrices have been computed again from these definitions, in exact integer arithmetic, by programs written independently of the program that generated the model, and compared with the model coefficient by coefficient: the 23 inequalities of Section 5 by the public program; the 269 matrices of Section 2, in the form reduced by symmetry, by one reconstruction; and the remaining 16095 constraints, namely the 941 inequalities $z(S) \ge 0$ and the 15154 constraints of Section 6, by another. Together these account for all 2074973 coefficients in the model. See Section 7 for the files.

1 The numbers $z(S)$ and two principles

For a word u and a code C put $X_u(C) = 1$ if $u \in C$, and $X_u(C) = 0$ otherwise. For a finite set S of words put $X_S = \prod_{u \in S} X_u$, with $X_\emptyset = 1$. Then $X_S X_{S'} = X_{S \cup S'}$, and X_S vanishes on every code if S is not compatible. A *polynomial* is a real linear combination $p = \sum_S p_S X_S$; its *degree* is the largest $|S|$ with $p_S \ne 0$. It is a function on the set of all codes.

Let C be a code. For a set S of at most four words put

$$z(S) = \frac{V}{18!} \sum_g X_S(gC),$$

where g runs over the permutations of [18]. Then $z(S)$ depends only on the orbit of S under these permutations, and

$$z(\emptyset) = V, \qquad z(\{u\}) = |C|, \qquad 0 \le z(S) \le V, \qquad z(S) = 0 \text{ if } S \text{ is not compatible.}$$

(For the second: the average of $X_u(gC)$ over g is $|C|/V$.)

(A) Let p be a polynomial of degree at most 4 such that $p(C') \ge 0$ for every code C'. Then $\sum_S p_S\, z(S) \ge 0$.

(B) Let h be a polynomial with $h(C') \ge 0$ for every code C', and let $f_1, \dots, f_r$ be polynomials such that every product hf_if_j has degree at most 4. Then the $r \times r$ matrix with entries $\sum_S (hf_if_j)_S\, z(S)$ is positive semidefinite.

Indeed, $\sum_S p_S z(S) = \frac{V}{18!} \sum_g p(gC)$, and every gC is a code. For (B), the quadratic form of the matrix at a vector a is $\frac{V}{18!} \sum_g h(gC) \big(\sum_i a_i f_i(gC)\big)^2 \ge 0$. Statement (B) is the principle of moment and localizing matrices [6].

The unknowns of the semidefinite program are the numbers $z(\omega)$, one for each orbit ω of compatible sets of one to four words; there are 941 of them. Every constraint below is an instance of (A) or (B). A principal submatrix of a positive semidefinite matrix, and its restriction to an invariant subspace of a group of symmetries, are again positive semidefinite; this is used to reduce the sizes of the matrices.

2 The constraints of the four-point bound

Let D be a compatible set of words, and consider the sets $S \supseteq D$ with $|D| + 2|S \setminus D| \le 4$. For two such sets S, S' we have $X_D X_S X_{S'} = X_{S \cup S'}$, of degree at most 4. By (B), with $h = X_D$ and the polynomials X_S, the matrix with entries $z(S \cup S')$ is positive semidefinite. These matrices, and the inequalities $z(S) \ge 0$, are the constraints of [8].

3 Local packings and their leaves

For disjoint sets $R, Z \subseteq [18]$ the *interval* $[R, Z]$ is the set of words u with $R \subseteq u$ and $u \cap Z = \emptyset$. Deleting the points of R and Z turns the words of a code in $[R, Z]$ into subsets of size $6 - |R|$ of a set of $18 - |R| - |Z|$ points, any two at distance at least 6.

In this section and the next two, $|R| = 2$ and $|Z| = b$. Let $Y = [18] \setminus (R \cup Z)$ and $m = |Y| = 16 - b$. For a code C, the sets $u \setminus R$ with $u \in C \cap [R, Z]$ are 4-subsets of Y, called *blocks*. Two blocks have at most one common point, because the two words have at most three. So the blocks form a *packing* P: every pair of points of Y lies in at most one block. For a 4-subset B of Y we write X_B for $X_{R \cup B}$; a set of 4-subsets of Y is compatible if it is a packing.

Assume that $m \ge 4$ and $m \equiv 1$ or $4 \pmod{12}$, that is, $b \in \{0, 3, 12\}$ and $m \in \{16, 13, 4\}$. Put

$$K = \frac{m(m-1)}{12}, \qquad N = |P|, \qquad t = K - N.$$

The *leave* L of P is the graph on Y whose edges are the pairs that lie in no block. A point in r blocks has degree $m - 1 - 3r$ in L, and L has $\binom{m}{2} - 6N = 6t$ edges. So t is a nonnegative integer, and every degree of L is divisible by 3.

Lemma 1. *If $t = 0$ then L has no edge. If $t = 1$ then L is a complete graph on four points, together with isolated points.*

Proof. Let $t = 1$. Then L has 6 edges and degree sum 12. Every non-isolated point has degree at least 3, so there are at most four of them; and 6 edges need at least four points. □

Let T be the number of triangles of L. Let I be the number of unordered pairs of blocks with exactly one common point, and Q the number of unordered triples of blocks that pairwise have exactly one common point and have no point common to all three.

Lemma 2. $T = \binom{m}{3} - (6m - 20)\,N + 9I - Q$.

Proof. A triple of points is a triangle of L if none of its three pairs lies in a block. By inclusion and exclusion, $T = \binom{m}{3} - \sigma_1 + \sigma_2 - \sigma_3$, where σ_i counts the choices of a triple and of i of its pairs that all lie in blocks. There are $6N$ pairs in blocks, each in $m-2$ triples, so $\sigma_1 = 6(m-2)N$. Two pairs of a triple lie either in the same block, which then contains the triple ($3 \cdot 4N$ choices), or in two blocks with exactly one common point, the common point of the two pairs ($3 \cdot 3$ choices for each such pair of blocks); so $\sigma_2 = 12N + 9I$. If all three pairs lie in blocks, the blocks are either the same ($4N$ triples) or three different ones, as counted by Q; two of the pairs cannot lie in one block and the third in another, since the first block contains the triple and the third pair would lie in two blocks. Three blocks as in the definition of Q meet pairwise in three different points, which form one such triple. So $\sigma_3 = 4N + Q$. □

Lemma 3. $F := T + 4t(t-2) \geq 0$ *for every packing* P.

Proof. If $t = 0$ then $T = 0$ and $F = 0$. If $t = 1$ then $T = 4$ by Lemma 1, and $F = 0$. If $t \geq 2$ both terms are nonnegative. □

By Lemma 2, and since $N = \sum_B X_B$ over all 4-subsets B of Y and $N^2 = N + 2\sum_{\{B,B'\}} X_B X_{B'}$, the function F is a polynomial of degree at most 3 in the indicators of the blocks, $F = \sum_S f(S) X_S$ over the compatible sets S of blocks, with

Set S of blocks	$f(S)$
empty	$\binom{m}{3} + 4K(K-2)$
one block	$32 - 8K - 6m$
two blocks	17 if they have one common point, 8 if they are disjoint
three blocks	-1 if they are as in the definition of Q, otherwise 0
four or more blocks	0

The following products are also nonnegative on every code, and have degree at most 4:

$$NF, \qquad tF = (K-N)F, \qquad H_{j,h}F,$$

where, for $0 \leq j \leq 2$ and $0 \leq h \leq b$, $H_{j,h}$ is the number of words u of the code with $|u \cap R| = j$ and $|u \cap Z| = h$. Since $H_{2,0} = N$, the pair $(j,h) = (2,0)$ is omitted. The coefficients of NF are $g(S) = |S|\, f(S) + \sum_{B \in S} f(S \setminus \{B\})$, because $X_B X_S$ is X_S if $B \in S$ and $X_{S \cup \{B\}}$ otherwise. The coefficients of tF are $K\, f(S) - g(S)$.

4 A lemma on graphs with twelve edges

A 4-*cycle* of a graph is a subgraph that is a cycle of length 4, induced or not.

Lemma 4. *Let* Γ *be a graph with* 12 *edges in which every degree is divisible by* 3. *Let* T *be its number of triangles and* C_4 *its number of* 4*-cycles. Then* $T + C_4 \geq 4$.

Proof. Delete the isolated points; let s points remain. Their degrees are at least 3 and add up to 24, so $s \leq 8$; hence the degrees are at most 7, that is, they are 3 or 6. If $s \leq 6$, all degrees are at most 5, hence equal to 3, and their sum is at most 18. So $s = 7$ or $s = 8$.

The case $s = 7$. One point x has degree 6 and the others have degree 3. Each of the other six points has two neighbours other than x, so there are six edges not on x, and each forms a triangle with x. Hence $T \geq 6$.

The case $s = 8$. Every degree is 3. For two points x, y let $c(x, y)$ be the number of their common neighbours. Counting paths of length two; triangles, by their edges; and 4-cycles, by their two pairs of opposite points, gives

$$\sum_{\{x,y\}} c(x,y) = 8\binom{3}{2} = 24, \qquad \sum_{\{x,y\} \text{ an edge}} c(x,y) = 3T, \qquad \sum_{\{x,y\}} \binom{c(x,y)}{2} = 2C_4.$$

There are $\binom{8}{2} - 12 = 16$ pairs that are not edges, and $\binom{c}{2} \geq c - 1$ for every integer $c \geq 0$. Adding this over the non-edges gives

$$2C_4 \ \geq \ (24 - 3T) - 16 = 8 - 3T.$$

So $C_4 \geq 4$ if $T = 0$, and $C_4 \geq 3$ if $T = 1$. If $T \geq 4$ there is nothing to prove. It remains to consider $T = 2$ and $T = 3$.

Two triangles with a common point have a common edge, since the common point has only three neighbours. Let xyp and xyq be triangles. Then x and y have no further neighbours. If p and q are adjacent, the four points carry four triangles. Otherwise p and q each have one neighbour among the other four points. These four points have degree sum 12, of which 2 is used by the edges to p and q; so they carry 5 edges, which form two triangles with a common edge. Again $T \geq 4$.

So if $T \leq 3$ the triangles are pairwise disjoint, and there are at most two of them, as there are only 8 points. Let $T = 2$, with triangles A and B and remaining points x, y. A point outside a triangle is adjacent to at most one of its points, since otherwise there is a third triangle. So x has at most one neighbour in A and at most one in B; having three neighbours, it is adjacent to y and to exactly one point of A, say a, and one of B. In the same way y is adjacent to exactly one point a' of A. Here $a' \neq a$, since a has only one neighbour outside A. Now x, a, a', y form a 4-cycle. There is another one through B. So $C_4 \geq 2$ and $T + C_4 \geq 4$. □

For example, the graph on the points $0, \ldots, 7$ with edges $\{i, i+1\}$ and $\{i, i+4\}$, taken modulo 8, has $T = 0$ and $C_4 = 4$. The case $s = 8$ was also confirmed by computer, by listing all 19355 graphs on 8 given points in which every degree is 3.

Lemma 5. *Let P be a packing as in Section 3, and let $\Sigma = T + C_4$ for its leave. Then*

$$H_2 := \Sigma + 5t^2 - 12t \ \geq 0, \qquad H_4 := 24\,\Sigma + t(t-3)(t-5)(5t-26) \ \geq 0.$$

Proof. If $t = 0$ the leave is empty and $\Sigma = 0$. If $t = 1$ the leave is a complete graph on four points, with 4 triangles and 3 4-cycles, so $\Sigma = 7$. If $t = 2$ the leave has 12 edges and all degrees divisible by 3, so $\Sigma \geq 4$ by Lemma 4. Always $\Sigma \geq 0$. The values of the two polynomials in t are

t	0	1	2	3	4	5
$5t^2 - 12t$	0	−7	−4	9	32	65
$t(t-3)(t-5)(5t-26)$	0	−168	−96	0	24	0

and both are positive for $t \geq 6$. □

For a pair e of points of Y let $Y_e = 1 - \sum_{B \ni e} X_B$. On a packing it is 1 if e is an edge of the leave and 0 otherwise. So T is the sum, over the triangles of the complete graph on Y, of the products of three Y_e, and C_4 is the corresponding sum over its 4-cycles, with four factors. Hence H_2 and H_4 are polynomials of degree at most 4 in the indicators of the blocks. Their coefficients are obtained from their values: for a compatible set S of at most four blocks,

$$c(S) = \sum_{S' \subseteq S} (-1)^{|S \setminus S'|}\, H(S'),$$

where $H(S')$ is the value of H_2 or H_4 on the packing S' in the same set Y. (As $H(P) = \sum_{S \subseteq P} c(S)$ for every packing P, this is Möbius inversion on the subsets of S; all subsets of a compatible set are compatible.)

5 From one interval to the numbers $z(S)$

Fix b, and let $p_{R,Z}$ be one of the polynomials F, NF, tF, $H_{j,h}F$, H_2, H_4 for the interval $[R,Z]$, written in the indicators X_u of the words. It is nonnegative on every code. So is the sum $p = \sum p_{R,Z}$ over all pairs (R,Z), and by (A)

$$\sum_S p_S\, z(S) \ \geq 0. \tag{1}$$

The coefficients p_S are found as follows. For a nonempty set S of words let $\iota(S)$ be the number of points in all words of S, and $\upsilon(S)$ the number of points in at least one. Then S lies in $\binom{\iota(S)}{2}\binom{18-\upsilon(S)}{b}$ intervals (here and below $\binom{x}{y} = 0$ if $y < 0$ or $y > x$): R consists of two points lying in all words of S, and Z of b points lying in none. Any two such choices differ by a permutation of the points that fixes every word of S, so the corresponding sets of blocks have the same coefficients. So for F, NF, tF, H_2 and H_4,

$$p_S = c(S)\binom{\iota(S)}{2}\binom{18-\upsilon(S)}{b}, \qquad p_\emptyset = c(\emptyset)\binom{18}{2}\binom{16}{b},$$

where c denotes the coefficients found in Sections 3 and 4.

For $H_{j,h}F$, a term arises from a word u counted by $H_{j,h}$ and a set S' of at most three words in the interval, with coefficient $f(S')$; here $u \notin S'$ because $(j,h) \neq (2,0)$. If S' is not empty, let A be the set of points in all words of S' and B the set of points in none. The number of intervals containing S' and with $|u \cap R| = j$, $|u \cap Z| = h$ is

$$\binom{|A \cap u|}{j}\binom{|A \setminus u|}{2-j}\binom{|B \cap u|}{h}\binom{|B \setminus u|}{b-h},$$

and if S' is empty it is $\binom{6}{j}\binom{12}{2-j}\binom{6-j}{h}\binom{10+j}{b-h}$. The coefficient of X_S is the sum, over the words $u \in S$, of $f(S \setminus \{u\})$ times this number.

Finally (1) is written in the unknowns $z(\omega)$. All sets S in an orbit ω have the same coefficient, so $z(\omega)$ gets the coefficient p_S times the size of ω. For a set of r distinct words, numbered in some order, let h_α be the number of points whose incidence with the r words is the pattern $\alpha \in \{0,1\}^r$, and let σ be the number of permutations of the r words that preserve all the numbers h_α. The permutations of the points that map the set to itself number $\sigma \prod_\alpha h_\alpha!$, so the orbit has $18!/(\sigma \prod_\alpha h_\alpha!)$ elements. The constant term is $p_\emptyset V$, as $z(\emptyset) = V$. Each inequality is divided by the greatest common divisor of its coefficients.

The 23 inequalities.

b	m	K	Inequalities
0	16	20	F, NF, tF; $H_{j,h}F$ for $(j,h) = (0,0), (1,0)$; H_2, H_4: seven inequalities
3	13	13	F, NF, tF; $H_{j,h}F$ for the eleven pairs $(j,h) \neq (2,0)$ with $0 \leq j \leq 2$, $0 \leq h \leq 3$; H_2, H_4: sixteen inequalities
12	4	1	none: all these polynomials vanish identically

For $m = 4$ there is at most one block, the leave is complete or empty, and F, H_2, H_4 are zero. The same construction for the complements of the words (weight 12) requires $|R| = 8$ to leave blocks of size 4, so that $m \leq 10$, and then $m = 4$.

6 The other constraints

In this section $[R,Z]$ is an arbitrary interval, $k = 6 - |R|$, $m = 18 - |R| - |Z|$, and N is the number of words of the code in $[R,Z]$. Then $\binom{N}{r} = \sum X_S$ over the r-sets S of words in $[R,Z]$,

so a polynomial of degree r in N is a polynomial of degree at most r in the indicators. The components of the vectors f below are polynomials of degree at most 1: the constant 1, counts of the words of the code in given classes with respect to a fixed partition of the points, and integer linear combinations of the indicators X_u that arise when the matrices are reduced by symmetry. In each family only products of degree at most 4 are used, and the polynomials or matrices for all intervals with the same $|R|$ and $|Z|$ are added, as in Section 5. An interval with $R = \emptyset$ consists of the words disjoint from Z, that is, of the words whose complements contain Z.

Sets that meet a code in at most one word. Let $r \geq 0$, let E be a set of $4 + 2r$ points and let $\mathcal{Q}$ be the set of words that have at least $4 + r$ points in E. Two words of $\mathcal{Q}$ have at least $2(4 + r) - (4 + 2r) = 4$ common points. So a code contains at most one word of $\mathcal{Q}$, and $q = 1 - \sum_{u \in \mathcal{Q}} X_u \geq 0$ on every code. The constraints are (A) for $X_D q$ with $|D| \leq 3$, and (B) for $h = X_u q$.

Intervals with pairwise disjoint blocks. If $\min(k, m - k) = 3$, the words in $[R, Z]$, or their complements in Y, are pairwise disjoint after deleting R. If moreover $9 \leq m < 12$, then $N \leq 3$, and for $j = 0, 1, 2, 3$

$$\sum_{r=j}^{3} (-1)^{r-j} \binom{r}{j} \binom{N}{r}$$

is 1 if $N = j$ and 0 otherwise. The constraints are (A) for the product of X_u and this polynomial.

Polynomials in N. Let $J(m, k) = 1$ if $\min(k, m - k) < 3$, and otherwise let $J(m, k)$ be the smaller of $\lfloor m\, J(m-1, k-1)/k \rfloor$ and $\lfloor m\, J(m-1, k)/(m-k) \rfloor$. The words on a given point, and the words not on it, form codes to which induction applies. Counting the pairs (word, point on it) and (word, point not on it) then gives $kN \leq m\, J(m-1, k-1)$ and $(m-k)N \leq m\, J(m-1, k)$, so $N \leq J(m, k)$. If π is a real polynomial of degree at most 4 with $\pi(i) \geq 0$ for all integers $0 \leq i \leq J(m, k)$, then $\pi(N) \geq 0$ on every code. The constraints are (A) for such $\pi(N)$.

Matrices from N. For an integer j, $(N - j)(N - j - 1) \geq 0$, and $N(J(m, k) - N) \geq 0$. The constraints are (B) with h one of these.

The known bound. Let $M = \sum_u X_u$ be the number of words of the code. Then $186 - M \geq 0$ on every code. The constraints are (A) for $X_D(186 - M)$ with $|D| \leq 3$, and (B) for $h = X_u(186 - M)$. The number 183 is not used anywhere.

7 The certificate

Number the unknowns $z_1, \ldots, z_{941}$, with $z_1 = z(\{u\}) = |C|$. Each constraint has the form: the symmetric matrix $F_\beta(z) = F_{\beta,0} + \sum_i z_i F_{\beta,i}$ is positive semidefinite, where the $F_{\beta,i}$ are integer matrices; a linear inequality is the case of a 1×1 matrix. There are 16387 such matrices, the largest of order 47. The certificate consists of rational positive semidefinite matrices Y_β of the same orders. With $\langle A, B \rangle = \sum_{i,j} A_{ij} B_{ij}$, put

$$c_0 = \sum_\beta \langle F_{\beta,0}, Y_\beta \rangle, \qquad \rho_i = \sum_\beta \langle F_{\beta,i}, Y_\beta \rangle + \begin{cases} 1 & (i = 1) \\ 0 & (i \neq 1). \end{cases}$$

For every code, $\langle F_\beta(z), Y_\beta \rangle \geq 0$ for all β, as both matrices are positive semidefinite. Adding, $c_0 - z_1 + \sum_i \rho_i z_i \geq 0$. Since $0 \leq z_i \leq V = 18564$,

$$|C| = z_1 \;\leq\; c_0 + 18564 \sum_i \max(\rho_i, 0) =: U.$$

The exact values are

$$\begin{aligned} c_0 &= 183.97846380536651084584137185081, \\ 18564 \textstyle\sum_i \max(\rho_i, 0) &= \;\; 0.00039641754292464531246590 46, \\ U &= 183.97886022290943549115383775541. \end{aligned}$$

These are rational numbers with denominators dividing 10^{29}, written out in full. As $|C|$ is an integer and $U < 184$, the Theorem follows. Bounding the terms $\rho_i z_i$ in this way is a standard device for exact certificates, see the appendix of [5].

Files and programs. All files are in the following repository, or attached to its releases.

Repository:	`github.com/wille614/cw-code-upper-bounds`
The matrices $F_{\beta,i}$, with their labels:	`A18_6_6_model.json.gz`
The matrices Y_β:	`A18_6_6_certificate.json.gz`

In a constraint, a constant a in a polynomial appears in $F_{\beta,0}$ as aV, in accordance with $z(\emptyset) = V$.

The command `python verify.py 18 6 6` computes the coefficients of the 23 inequalities of Section 5 again, from direct counts of triangles and 4-cycles and of intervals, and compares them with the model; it verifies that every Y_β is positive semidefinite, by exact elimination in rational arithmetic; and it computes c_0, all ρ_i and U exactly.

The directory `reconstruction/A18_6_6/` contains the specification of the labels and two sets of programs with their reports. One computes again the 941 inequalities $z(S) \geq 0$ and the 15154 constraints of Section 6, the other the 269 matrices of Section 2 in the form reduced by symmetry. Each compares its result with the model, coefficient by coefficient.

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